Consultant: PARSONS	Client: CROWN PRINCE PRIVATE AFFAIRS	Contractor: MBL MODERN BUILDING LEADERS (MBL) Co. LLC
Project Name: L01, Riyadh, Kingdom of Saudi Arabia		

INSPECTION AND TEST PLAN

FOR

CONCRETE WORK

DOCUMENT NUMBER: L01-MBLC-CON-CIV-ITP-000164-R01						
No.	Remarks	Rev.	Prepared By:	Checked By:	Approved By:	Date
01	ITP Submittal	0	S. A. Basheer (QA/QC Engineer)	Anil Narayana (QHSE Director)	Khalil Atieh (Project Director)	21/09/2020

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1. Purpose

The purpose of the Inspection and Test Plan (ITP) is to ensure that the CONCRETE WORK is accordance with the requirements of L01 Project.

2. Scope

This ITP shall be implemented for the Concrete Works in the L01 Project. This ITP covers all the stages of the Concrete works.

3. Definitions

Test	Test means determination of one or more characteristics according to a procedure.
Activity	It relates to the sequential items of work process.
Verification	It is confirmation, through the provision of objective evidence that specified requirements have been fulfilled.
Verification Documentation/Test Data	The record raise to verify that a test or inspection has been undertaken in accordance with the specified requirements. This document will detailthe pass or failure of the test or inspection against the specified acceptance criteria.
Inspections	Inspection means examination, verification, testing, measurements and documentation to confirm item such as equipment, material, components, parts, assemblies, systems and structures resulting from manufacturing or construction processes conform to the Client specification and or applicable international codes and standards.
Receiving Inspection	Receiving inspection means, inspections to verify conformance to requirements of items/materials when taking delivery of items/ materials at a designated location.
In-process Inspection	In-process Inspection means, inspection of an action to verify conformance to requirements of the activity in-processing items/materials.
Final Inspection	Final Inspection means, inspection to verify conformance of completed products of activity to specified requirements.
Inspector	Inspector means, a qualified person who performs inspection activities to verify conformance to specific requirements.

Hold Point (H)	An Inspection point defined in an appropriate document in which an approved contractors/subcontractor's QC inspectors are obliged to execute the inspection/test to verify and ensure that the inspectable item or activity is in compliance with Client requirements. prior to submitting Work Inspection Request (WIR) addressed to the Consultant Engineer 24 hours before the date of inspection. And succeeding activity shall not commence unless the current activity is inspected and accepted by the Engineer.
Witness Point (W)	An inspection point, defined in an appropriate document, beyond which an activity can proceed after notifying the designated Client/Consultant Engineer. The contractor QC Inspector has full responsibility for performing detailed inspection during the work activity.
Review Point (R)	A review point, defined in an appropriate document, is the point at which a work activity cannot proceed without first ensuring the review and obtain approval of applicable documents or personnel qualifications by a designated authority
Surveillance Point (S)	The party involved is requested to perform daily or random inspections, monitoring the relevant results, on construction activities/processes as indicated in the description of item to be inspected.
Inspection Test Plan	ITP means, a document that provides detailed direction for the quality verification and inspection activities performed by contractor Quality Inspector and nominated supplier/subcontractor. Inspection Test Plan are controlling documents that specify what inspections and test are required; what the standard of acceptance are, how by whom, and at what frequency and sequence the inspection are to be performed.
Project Quality Plan (PQP)	PQP means, a record that provides a documented record of completed inspection / test results.

Record

A document stating results achieved or providing evidence of activities performed.

4. Records

Approved ITPs, copies of all completed verification documents and associated audit, surveillance and inspection reports shall be maintained as records.

Additionally, those documents identified as project records for self-certification purposes shall be collated and maintained separately

5. Appendices

5.1 Appendix 1

INSPECTION & TEST PLAN

INSPECTION AND TEST PLAN FOR CONCRETE WORK			ITP Reference:		REVISION:
Activity:	CONCRETE WORK (General, Pre & post pouring inspection)	Discipline: CIVIL		DRAWING Nos. OR DOCUMENT REFERENCES:	
Location:	L-01			Method of Statements -L01-MBLC-CON-CIV-MES-0047-R00/ L01-SAPL-CON-MES-TRN-0166 Project Quality plan- L01-MBLC-MGT-GEN-PQP-0003-02 Concrete work specs: L01-MBLC-D01-SPE-STR-0067-R00	

SL No	Activity	Acceptance Criteria	Verification Activity and Documentation	Responsible	Frequency If Applicable	CONTROL POINTS				RECORD
						SUB	MBL	DAR	3rd party	
1	General									
1.1	Permission to commence construction Approved MST	Approved MST	Document review	Project Director/ QC Engineer	Prior to starting work	I/P	H	R		Drawings/ documents/ logs
1.2	Compilation of Record Packages including Close out of NCRs.	All relevant documents	All activities should be mention in Work inspection request/Test reports	Project Engineer/QC Engineer	Only on completion completeness of package & all NCRs closed out.	I/P	H	R		Work Inspection Request
1.3	Approval of Third-Party Testing Lab	Company Approval Letter	Company Profile, Lab accreditations, etc.,	Project Engineer/QC Engineer	Prior to starting work	I/P	H	R		Company Approval Letter
1.4	Approval of Concrete Mix Design	Approved Material by SAPL Consultant	Approved Design mix documents	Project Engineer/QC Engineer	Prior to starting work	I/P	H	R		Material Approval request
1.5	Carry out inspection & test of raw materials Approval Cement	Approved Material by SAPL Consultant	Material Approval Request	Project Engineer/QC Engineer	Prior to starting work	I/P	R	R	H	Material Approval request
1.6	Approval of Admixtures	Approved Material by SAPL Consultant	Material Approval Request	Project Engineer/QC Engineer	Prior to starting work	I/P	R	R	H	Material Test certificates
1.7	Approval of Aggregates	Approved Material by SAPL Consultant	Material Approval Request	Project Engineer/QC Engineer	Prior to starting work	I/P	R	R	H	Material Test Certificates
1.8	Trial Mix	As per Contract Standards	As per Contract Stand	Project Engineer/QC Engineer	Prior to starting work	I/P	H	H	H	Trial Mix reports
1.9	Compressive Strength of Trial Mix Samples including Standard Deviation of strength Values	As per Contract Standards	As per Contract Standards	Project Engineer/QC Engineer	Prior to starting work	I/P	W	W	H	Third party lab test reports
1.10	Approval of Reinforcement steel	Approved Material by SAPL Consultant	Material Approval Request	Project Engineer/QC Engineer	Prior to starting work	I/P	W	R	H	As record
1.11	Approval of water Source supply	As per Contract Standards	As per contract standards	Project Engineer/QC Engineer	Prior to starting work	I/P	R	R	H	Third party lab test reports
1.12	Approval of Embedded Items /Anchor Blots	Approved Material by SAPL Consultant	Material Approval Request	Project Engineer/QC Engineer	Prior to starting work	I/P	R	R	H	Material Approval request
1.13	Approval of form material	Approved Material by SAPL Consultant	Material Approval Request	Project Engineer/QC Engineer	Prior to starting work	I/P	R	R		Material Approval Request

SL No	Activity	Acceptance Criteria	Verification Activity and Documentation	Responsible	Frequency If Applicable	CONTROL POINTS				Record
						SUB	MC	CONSULTANT	3rd party	
2	**CONCRETE PRE-POUR INSPECTION**									
2.1	(Blinding/ Reinforced) Concrete	(C14/C40) as per item and approved mix design according to project Spec.s	Tests / delivery report / WIR / visual inspection	Project Engineer / QC Engineer	Prior to raft slab	I/P	W	W		As record
2.2	Fixing of reinforcement steel	to be installed as per standards and approved Drawings	Approved Method of Statement / MTS / WIR / visual inspection	Project Engineer / QC Engineer	Prior to formwork	I/P	H	H		As record
2.3	Fixing of Anchor Bolts / Embedded Items	Material approval, installation	Approved Method of Statement/ WIR	Project Engineer / QC Engineer	During reinforcement and formwork installation	I/P	H	H		As record
2.4	Fixing & Alignment of Formwork	Material approval, installation	Approved Method of Statement/ WIR	Project Engineer / QC Engineer	After reinforcement steel installation	I/P	H	H		As record
2.5	Survey Works (Co-ordinates & Elevation)	As per approved DWG	Surveyor report, WIR	Project Engineer / Surveyor	Prior to every concrete work	I/P	H	H		Survey Report shall be prepared
2.6	Concrete pouring	As per prior activities inspections (steel, formwork and survey)	Approved Method of Statement/ WIR	Project Engineer / QC Engineer	Prior to every concrete pouring	I/P	H	H		Clearance should be signed by all trades
3	**CONCRETE POURING INSPECTION**									
3.1	Concrete Mixing Duration, Time of Transportation,	As per Standards, Duration is 90 min max after starting mixing concrete.	Delivery note	Project Engineer / QC Engineer	Each mixing truck delivery.	I/P	W	R	H	Delivery Note
3.2	Slump Test	As per Standard	Test report	3rd party Engineer / QC Engineer	Each delivery	I/P	H	W	H	As record
	Air content	Shouldn't exceeds 2.0%	Test report	Project Engineer / QC Engineer	Each delivery	I/P			H	As record
3.3	Temperature Check (Concrete & Ambient)	30 C degree max	Test report	Project Engineer / QC Engineer	Each delivery	I/P	H	S	H	As record
	Compressive Strength of Cubes (7day & 28 day)	As per Standards	Test report	Project Engineer / QC Engineer	Every 150 M3	I/P	W	S	H	As record
3.4	Concrete Sampling (Cylinder)	As per Standards	Test report	Project Engineer / QC Engineer	Every 25 M3	I/P	W	S	H	As record
3.5	Concrete Placing	As per Standards	Tests reports reinforcement/MST/ material approval	Project Engineer / QC Engineer	Every time	I/P	W	S		As record

SL No	Activity	Acceptance Criteria	Verification Activity and Documentation	Responsible	Frequency	CONTROL POINTS				Record
						SUB	MC	CONSULTANT	3rd party	
4.0	Concrete post pouring inspection									
4.1	Ensuring protection and curing of concrete for minimum period of 7 days	Shall be complying to ASTM C1602/C1602M	Visual inspection	Project Engineer / QC Engineer	After pouring	I/P	W	S		
4.2	Ensure Concrete repair prior to cover up	As per Standards	visual inspection/WIR	Project Engineer / QC Engineer	Whenever needed	I	H	H		As Record

S = Surveillance – Random Physical Check Record, Activities and Process,
R = In-process Inspection – Work Continues – After Ensuring the Review and Obtain Approval of Applicable Documents or Personnel Qualification by Authority.
W = Witness Inspection – Work Continues – After Notifying the Client/Consultant Engineer
H = Hold Point – Ensure Item/Activity id Inspection by QC inspected prior to raising WIR. Work succeeding cannot proceed until current activity is inspected and accepted by Client/Consultant Engineer
A = Approval (Work Content Only), I/P- INITIATIVE/PERFORM, SAPL- SAUDI ARABIAN PERSONS LIMITED

5.2 Appendix 2
CHECK LIST FOR CONCRETE WORK

CHECKLIST					Ref No: Date:16/06/2020
Location Description	L-01				
Examination Items	**MBL QA/QC**		**DAR**		
	Satisfied	Unsatisfied	Satisfied	Unsatisfied	Remark
Anti-termite treatment					
Form work dimensions, tightness, rigidity, cleanliness					
Scaffolding					
Reinforcement Bar size, numbers, spacing, laps, cleanliness					
Water Stoppers installed at construction joints					
Existing Concrete cleaning					
Surveying, level and coordinates					
Method of pouring & Vibration					
Arrangement for curing					

Comments :				
Site Engineer			**QC Engineer**	**Consultant Engineer**
Name:			Name:	Name:
Signature:			Signature:	Signature:
Date:			Date:	Date: